

Multi-Branch Enterprise Network Design

A Cisco Packet Tracer WAN Simulation Project Report

Done by

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1. Introduction

This report documents the design and implementation of a multi-branch enterprise network, simulated in Cisco Packet Tracer. The company operates a central Headquarters (HQ) in Dhaka and two branch offices located in Chattogram and Sylhet. Although geographically separated, all three sites are interconnected across a Wide Area Network (WAN) so that they function as a single, unified corporate network.

Each site maintains its own local network (LAN) with departmental VLANs and local DHCP, while a dynamic routing protocol (OSPF) ensures that every site automatically learns how to reach every other site. This design is scalable: additional branches can be added later with minimal reconfiguration.

2. Project Objectives

- Connect a Headquarters and two branch offices into one enterprise-wide network.
- Assign non-overlapping, hierarchical IP addressing across all sites.
- Establish WAN links between HQ and each branch using point-to-point connections.
- Configure OSPF dynamic routing so all sites share reachability information automatically.
- Verify end-to-end connectivity between users at different physical sites.

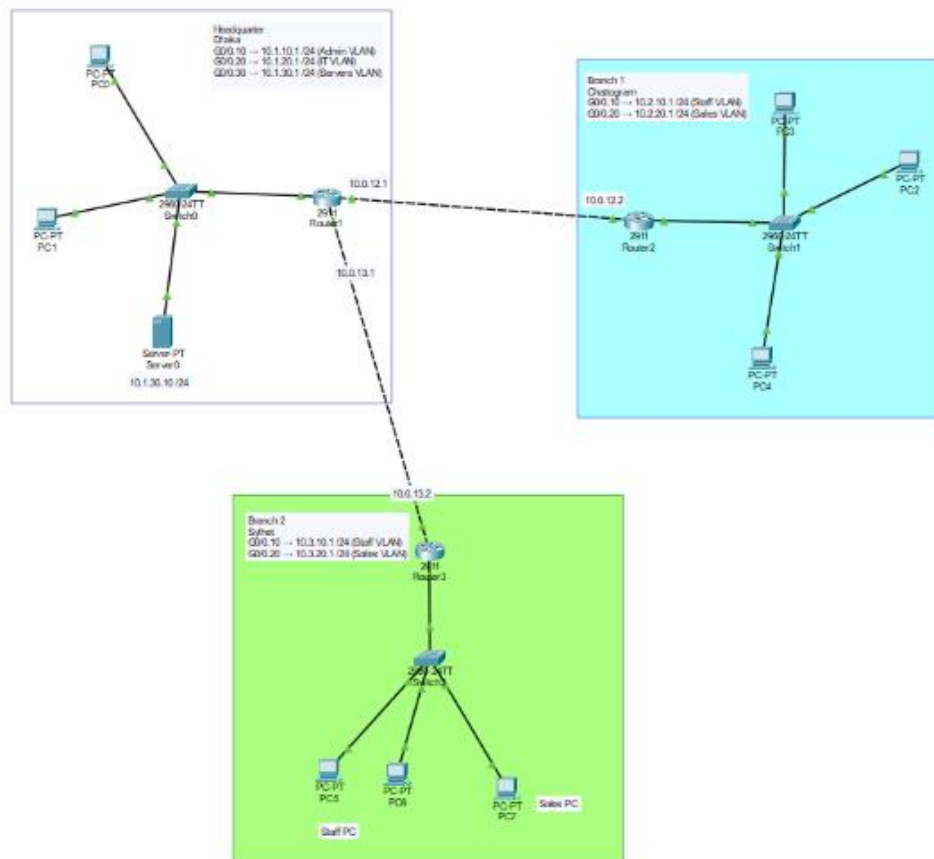


Figure 1 Multi-Branch Enterprise Network Design

3. WAN Topology Overview

The network uses a hub-and-spoke topology. HQ acts as the central hub, and each branch office connects to HQ over a dedicated WAN link (the spokes). All inter-branch traffic is routed through HQ. This is the most common enterprise WAN model because it is simple to manage, cost-effective, and easy to scale.

Logical layout:

- HQ (Dhaka): central hub. Hosts corporate servers and IT administration.
- Branch 1 (Chattogram): spoke, connected to HQ via WAN link 1.
- Branch 2 (Sylhet): spoke, connected to HQ via WAN link 2.

Equipment per site:

- 1 × Router (Cisco 2911) : LAN gateway, WAN link, and OSPF routing.
- 1 or more × Switch (Cisco 2960) : connects local end devices and VLANs.
- PCs and (at HQ) a server for corporate services.

4. Site & LAN Addressing Plan

The 10.0.0.0/8 private range is divided so each site receives its own /16 block. This keeps addressing organized, prevents overlap, and makes routing tables clean and summarizable.

Site	Role	LAN Block	VLAN / Subnet	Gateway
HQ (Dhaka)	Headquarters	10.1.0.0/16	Admin — 10.1.10.0/24	10.1.10.1
HQ (Dhaka)	Headquarters	10.1.0.0/16	IT — 10.1.20.0/24	10.1.20.1
HQ (Dhaka)	Headquarters	10.1.0.0/16	Servers — 10.1.30.0/24	10.1.30.1
Branch 1 (Ctg)	Branch office	10.2.0.0/16	Staff — 10.2.10.0/24	10.2.10.1
Branch 1 (Ctg)	Branch office	10.2.0.0/16	Sales — 10.2.20.0/24	10.2.20.1
Branch 2 (Syl)	Branch office	10.3.0.0/16	Staff — 10.3.10.0/24	10.3.10.1
Branch 2 (Syl)	Branch office	10.3.0.0/16	Sales — 10.3.20.0/24	10.3.20.1

5. WAN Link Addressing

Each WAN link between two routers is a point-to-point connection using a /30 subnet, which provides exactly two usable host addresses — one for each router. A separate 10.0.x.x range is reserved for these links to keep them distinct from LAN traffic.

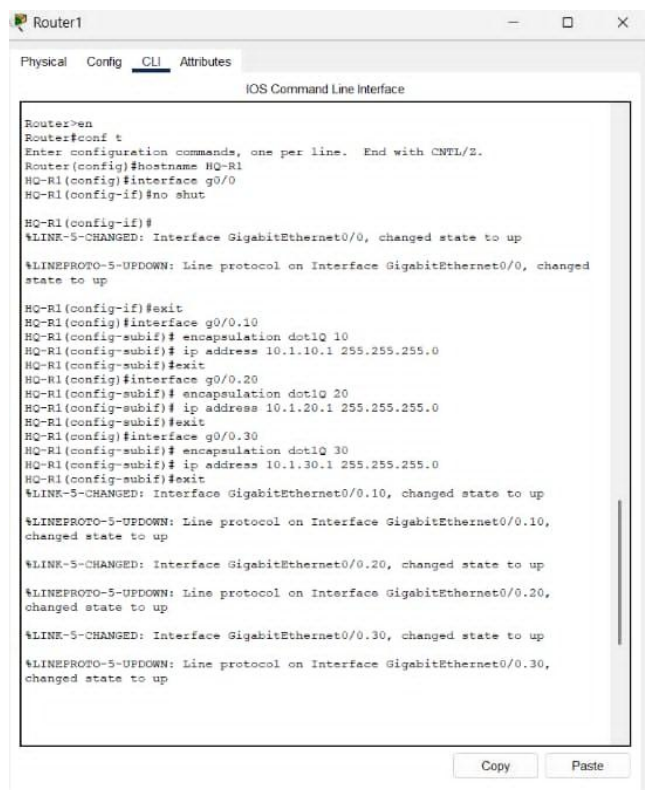
WAN Link	Subnet	HQ Interface	Branch Interface
HQ ↔ Branch 1	10.0.12.0/30	G0/1 — 10.0.12.1	G0/1 — 10.0.12.2
HQ ↔ Branch 2	10.0.13.0/30	G0/2 — 10.0.13.1	G0/1 — 10.0.13.2

6. Dynamic Routing with OSPF

OSPF (Open Shortest Path First) is used as the routing protocol. Each router advertises its local LAN networks and its WAN links into OSPF Area 0. OSPF then floods this information so every router builds a complete map of the enterprise and automatically calculates the best path to every subnet. Unlike static routing, no manual route entries are needed, and adding a new branch requires only enabling OSPF on the new router.

7. Key Configuration Commands

7.1 HQ Router (Hub) — HQ router configuration, sub-interface per VLAN, WAN Interfaces & OSPF



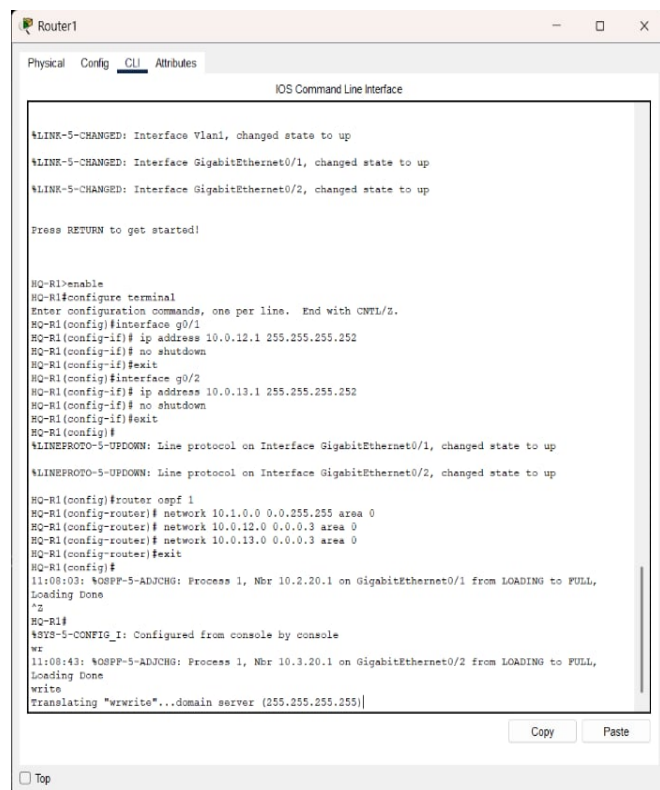
```
Router1>en
Router1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#hostname HQ-R1
HQ-R1(config)#interface g0/0
HQ-R1(config-if)#no shut

HQ-R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

HQ-R1(config-if)#exit
HQ-R1(config)#interface g0/0.10
HQ-R1(config-subif)# encapsulation dot1Q 10
HQ-R1(config-subif)# ip address 10.1.10.1 255.255.255.0
HQ-R1(config-subif)#exit
HQ-R1(config)#interface g0/0.20
HQ-R1(config-subif)# encapsulation dot1Q 20
HQ-R1(config-subif)# ip address 10.1.20.1 255.255.255.0
HQ-R1(config-subif)#exit
HQ-R1(config)#interface g0/0.30
HQ-R1(config-subif)# encapsulation dot1Q 30
HQ-R1(config-subif)# ip address 10.1.30.1 255.255.255.0
HQ-R1(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/0.30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
```



```
Router1>en
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router1(config)#interface g0/1
Router1(config-if)# ip address 10.0.12.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)#exit
Router1(config)#interface g0/2
Router1(config-if)# ip address 10.0.13.1 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)#exit
Router1(config)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router1(config)#router ospf 1
Router1(config-router)# network 10.1.0.0 0.0.255.255 area 0
Router1(config-router)# network 10.0.12.0 0.0.0.3 area 0
Router1(config-router)# network 10.0.13.0 0.0.0.3 area 0
Router1(config-router)#exit
Router1(config)#
11:08:43: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.20.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
^Z
Router1#
SYS-5-CONFIG_I: Configured from console by console
vr
11:08:43: %OSPF-5-ADJCHG: Process 1, Nbr 10.3.20.1 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
write
Translating "write"...domain server (255.255.255.255)
```

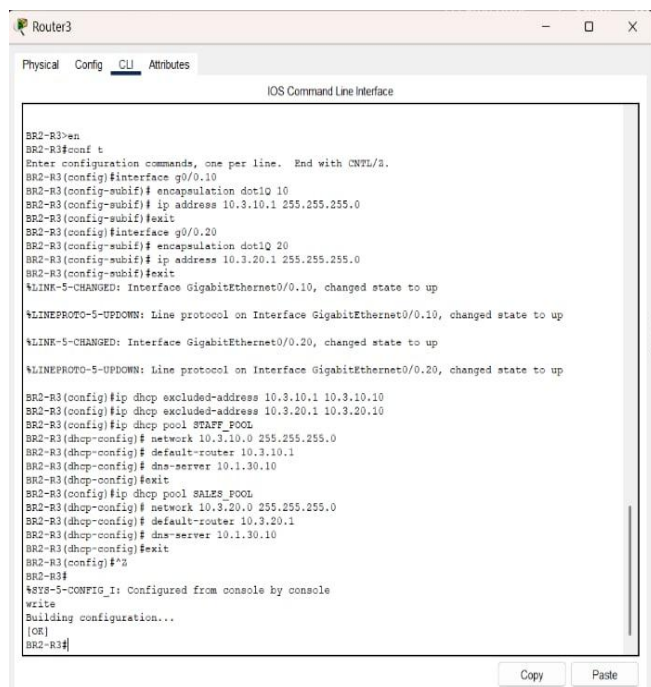
7.2 Branch 1 Router (Chattogram) — WAN & OSPF



The screenshot shows the CLI of Router2 with the following configuration commands and output:

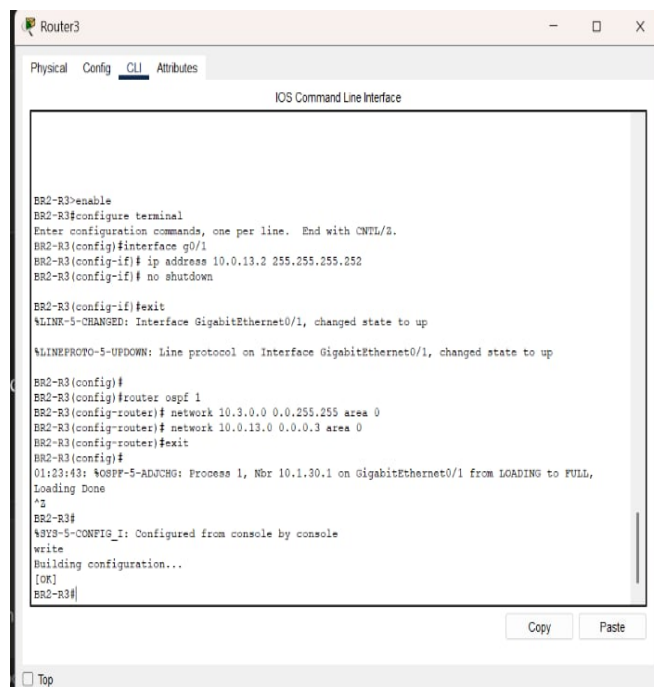
```
Router2>enable
Router2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router2(config)#hostname BR1-R1
BR1-R1(config)#interface g0/0
BR1-R1(config-if)# no shutdown
BR1-R1(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
BR1-R1(config)#interface g0/0.10
BR1-R1(config-subif)# encapsulation dot1Q 10
BR1-R1(config-subif)# ip address 10.2.10.1 255.255.255.0
BR1-R1(config-subif)#exit
BR1-R1(config)#interface g0/0.20
BR1-R1(config-subif)# encapsulation dot1Q 20
BR1-R1(config-subif)# ip address 10.2.20.1 255.255.255.0
BR1-R1(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
BR1-R1(config)#ip dhcp excluded-address 10.2.10.1 10.2.10.10
BR1-R1(config)#ip dhcp excluded-address 10.2.20.1 10.2.20.10
BR1-R1(config)#ip dhcp pool STAFF_POOL
BR1-R1(dhcp-config)# network 10.2.10.0 255.255.255.0
BR1-R1(dhcp-config)# default-router 10.2.10.1
BR1-R1(dhcp-config)# dns-server 10.1.30.10
BR1-R1(dhcp-config)#exit
BR1-R1(config)#ip dhcp pool SALES_POOL
BR1-R1(dhcp-config)# network 10.2.20.0 255.255.255.0
BR1-R1(dhcp-config)# default-router 10.2.20.1
BR1-R1(dhcp-config)# dns-server 10.1.30.10
BR1-R1(dhcp-config)#exit
```

7.3 Branch 2 Router (Sylhet) — WAN & OSPF



The screenshot shows the CLI of Router3 with the following configuration commands and output:

```
BR2-R3>en
BR2-R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
BR2-R3(config)#interface g0/0.10
BR2-R3(config-subif)# encapsulation dot1Q 10
BR2-R3(config-subif)# ip address 10.3.10.1 255.255.255.0
BR2-R3(config-subif)#exit
BR2-R3(config)#interface g0/0.20
BR2-R3(config-subif)# encapsulation dot1Q 20
BR2-R3(config-subif)# ip address 10.3.20.1 255.255.255.0
BR2-R3(config-subif)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
BR2-R3(config)#ip dhcp excluded-address 10.3.10.1 10.3.10.10
BR2-R3(config)#ip dhcp excluded-address 10.3.20.1 10.3.20.10
BR2-R3(config)#ip dhcp pool STAFF_POOL
BR2-R3(dhcp-config)# network 10.3.10.0 255.255.255.0
BR2-R3(dhcp-config)# default-router 10.3.10.1
BR2-R3(dhcp-config)# dns-server 10.1.30.10
BR2-R3(dhcp-config)#exit
BR2-R3(config)#ip dhcp pool SALES_POOL
BR2-R3(dhcp-config)# network 10.3.20.0 255.255.255.0
BR2-R3(dhcp-config)# default-router 10.3.20.1
BR2-R3(dhcp-config)# dns-server 10.1.30.10
BR2-R3(dhcp-config)#exit
BR2-R3(config)#%
BR2-R3#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
BR2-R3#
```



The screenshot shows the CLI of Router3 with the following configuration commands and output:

```
BR2-R3>enable
BR2-R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BR2-R3(config)#interface g0/1
BR2-R3(config-if)# ip address 10.0.13.2 255.255.255.252
BR2-R3(config-if)# no shutdown
BR2-R3(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
BR2-R3(config)#
BR2-R3(config)#router ospf 1
BR2-R3(config-router)# network 10.3.0.0 0.0.255.255 area 0
BR2-R3(config-router)# network 10.0.13.0 0.0.0.3 area 0
BR2-R3(config-router)#exit
01:23:43: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.30.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done
BR2-R3#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
BR2-R3#
```

8. Testing & Verification

The following commands confirm that all sites form one working network:

- show ip ospf neighbor –

```
HQ-R1>
HQ-R1>show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.2.20.1        1     FULL/DR         00:00:37    10.0.12.2      GigabitEthernet0/1
10.3.20.1        1     FULL/DR         00:00:37    10.0.13.2      GigabitEthernet0/2
HQ-R1>
```

- show ip route ospf –

```
HQ-R1>
HQ-R1>show ip route ospf
      10.0.0.0/8 is variably subnetted, 14 subnets, 3 masks
O       10.2.10.0 [110/2] via 10.0.12.2, 00:48:10, GigabitEthernet0/1
O       10.2.20.0 [110/2] via 10.0.12.2, 00:48:10, GigabitEthernet0/1
O       10.3.10.0 [110/2] via 10.0.13.2, 00:48:10, GigabitEthernet0/2
O       10.3.20.0 [110/2] via 10.0.13.2, 00:48:10, GigabitEthernet0/2
HQ-R1>
```

```
BR1-R1>
BR1-R1>show ip route ospf
      10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks
O       10.0.13.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.10.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.20.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.1.30.0 [110/2] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.3.10.0 [110/3] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
O       10.3.20.0 [110/3] via 10.0.12.1, 00:49:23, GigabitEthernet0/1
BR1-R1>
```

- show ip interface brief - confirms WAN interfaces are up/up.

```
BR1-R1>
BR1-R1>show ip interface brief

Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES unset   up          up
GigabitEthernet0/0.10 10.2.10.1      YES manual  up          up
GigabitEthernet0/0.20 10.2.20.1      YES manual  up          up
GigabitEthernet0/1    10.0.12.2      YES manual  up          up
GigabitEthernet0/2    unassigned      YES unset   administratively down down
Vlan1               unassigned      YES unset   administratively down down
BR1-R1>
```

```
BR2-R3>
BR2-R3>show ip interface brief

Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES unset   up          up
GigabitEthernet0/0.10 10.3.10.1      YES manual  up          up
GigabitEthernet0/0.20 10.3.20.1      YES manual  up          up
GigabitEthernet0/1    10.0.13.2      YES manual  up          up
GigabitEthernet0/2    unassigned      YES unset   administratively down down
Vlan1               unassigned      YES unset   administratively down down
BR2-R3>
```

- Cross-site ping –

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.3.20.11

Pinging 10.3.20.11 with 32 bytes of data:

Request timed out.
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126

Ping statistics for 10.3.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.3.20.11

Pinging 10.3.20.11 with 32 bytes of data:

Reply from 10.3.20.11: bytes=32 time<1ms TTL=126
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126
Reply from 10.3.20.11: bytes=32 time<1ms TTL=126

Ping statistics for 10.3.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|

```

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.2.20.11

Pinging 10.2.20.11 with 32 bytes of data:

Request timed out.
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125

Ping statistics for 10.2.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.2.20.11

Pinging 10.2.20.11 with 32 bytes of data:

Reply from 10.2.20.11: bytes=32 time<1ms TTL=125
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125
Reply from 10.2.20.11: bytes=32 time<1ms TTL=125

Ping statistics for 10.2.20.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

- Ping to HQ server-

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.30.10

Pinging 10.1.30.10 with 32 bytes of data:

Request timed out.
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.1.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.1.30.10

Pinging 10.1.30.10 with 32 bytes of data:

Reply from 10.1.30.10: bytes=32 time<1ms TTL=126
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126
Reply from 10.1.30.10: bytes=32 time<1ms TTL=126

Ping statistics for 10.1.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

9. Conclusion

The project demonstrates a scalable multi-branch enterprise network in which three geographically separate sites operate as a single unified network. Hierarchical /16 addressing keeps each site organized, point-to-point WAN links interconnect the offices, and OSPF dynamic routing automatically maintains full reachability between all sites. The design scales cleanly: a new branch can be integrated simply by adding a WAN link to HQ and enabling OSPF. Future enhancements could include inter-site access-control policies, WAN link redundancy, and per-department DHCP at each branch.